

Monitoring & Maintenance Guide

Health Checks

Automated Health Monitoring Script

Save as `/opt/fel/health_check.sh` and run via cron every 5 minutes:

```
#!/bin/bash
ALERT_EMAIL="ops@finalevolutiongroup.com"
SLACK_WEBHOOK="https://hooks.slack.com/services/xxx"

check_service() {
    local name=$1 url=$2 expected=$3
    local status=$(curl -s -o /dev/null -w "%{http_code}" --max-time 5 "$url")
    if [ "$status" != "$expected" ]; then
        echo "ALERT: $name returned $status (expected $expected)"
        # Send alert via Slack/email
        curl -s -X POST "$SLACK_WEBHOOK" -d "{\"text\": \"🚨 $name is DOWN (HTTP $status)\"}" > /dev/null
    fi
}

# Check services
check_service "Signalling" "http://localhost:8888/healthz" "200"
check_service "Website" "https://finalevolutiongroup.com" "200"
check_service "Web App" "https://app.finalevolutiongroup.com" "200"

# Check GPU utilization
GPU_UTIL=$(nvidia-smi --query-gpu=utilization.gpu --format=csv,noheader,nounits 2>/dev/null || echo "N/A")
if [ "$GPU_UTIL" != "N/A" ] && [ "$GPU_UTIL" -gt 95 ]; then
    echo "WARNING: GPU utilization at ${GPU_UTIL}%"
fi

# Check disk space
DISK_USED=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%')
if [ "$DISK_USED" -gt 85 ]; then
    echo "WARNING: Disk usage at ${DISK_USED}%"
fi
```

Cron Setup

```
# Add to crontab
*/5 * * * * /opt/fel/health_check.sh >> /var/log/fel/health_check.log 2>&1
```

Metrics to Monitor

Metric	Target	Alert Threshold
Signalling uptime	99.9%	<99%
WebRTC latency	<30ms	>100ms
Stream quality	1080p@60fps	<720p or <30fps
Active sessions	Track	>50 concurrent
GPU utilization	40-80%	>95%
CPU utilization	<70%	>90%
Memory usage	<80%	>90%
Disk usage	<70%	>85%
SSL cert expiry	>30 days	<14 days
API response time	<200ms	>1000ms

Log Management

Log Locations

```

/home/ubuntu/rork-final-evolution-lab/logs/
streaming/signalling/logs/
streaming/frontend/logs/
/var/log/nginx/
/var/log/coturn/
# Pipeline logs
# Signalling server
# Frontend
# Nginx access/error
# TURN server

```

Log Rotation

```

# /etc/logrotate.d/fel
/home/ubuntu/rork-final-evolution-lab/logs/*.log {
    daily
    missingok
    rotate 14
    compress
    delaycompress
    notifempty
}

```

Maintenance Procedures

Weekly

- [] Review error logs
- [] Check SSL certificate expiry dates
- [] Verify backup integrity
- [] Review active session metrics

Monthly

- [] Update OS packages: `sudo apt update && sudo apt upgrade`
- [] Update Node.js dependencies: `npm audit fix`
- [] Review Cloudflare analytics
- [] Check GPU driver updates
- [] Run full integration test suite

Quarterly

- [] Renew SSL certificates (if not auto-renewed)
- [] Review and update security policies
- [] Performance benchmarking
- [] Disaster recovery drill

Backup Strategy

```
# Daily backup script
#!/bin/bash
BACKUP_DIR="/backups/fel/$(date +%Y-%m-%d)"
mkdir -p $BACKUP_DIR

# Source code
tar -czf $BACKUP_DIR/source.tar.gz \
  --exclude='node_modules' --exclude='.git' --exclude='Builds' \
  /home/ubuntu/rork-final-evolution-lab/

# Configuration
cp /etc/nginx/sites-available/finalevolution* $BACKUP_DIR/
cp /home/ubuntu/rork-final-evolution-lab/.env $BACKUP_DIR/

# Cleanup: keep 30 days
find /backups/fel/ -maxdepth 1 -mtime +30 -type d -exec rm -rf {} +
```

Scaling Guide

Horizontal Scaling (More Users)

1. Add more GPU instances behind load balancer
2. Use shared signalling server with session affinity
3. Configure CoTURN cluster with shared secret

Vertical Scaling (Better Quality)

1. Upgrade GPU (T4 → A10G → A100)
2. Increase streaming resolution (1080p → 4K)
3. Increase target FPS (60 → 120)